

Model system

$$\text{SDE: } \frac{dx}{dt} = A \frac{dx}{dt} + Bu + \varepsilon$$

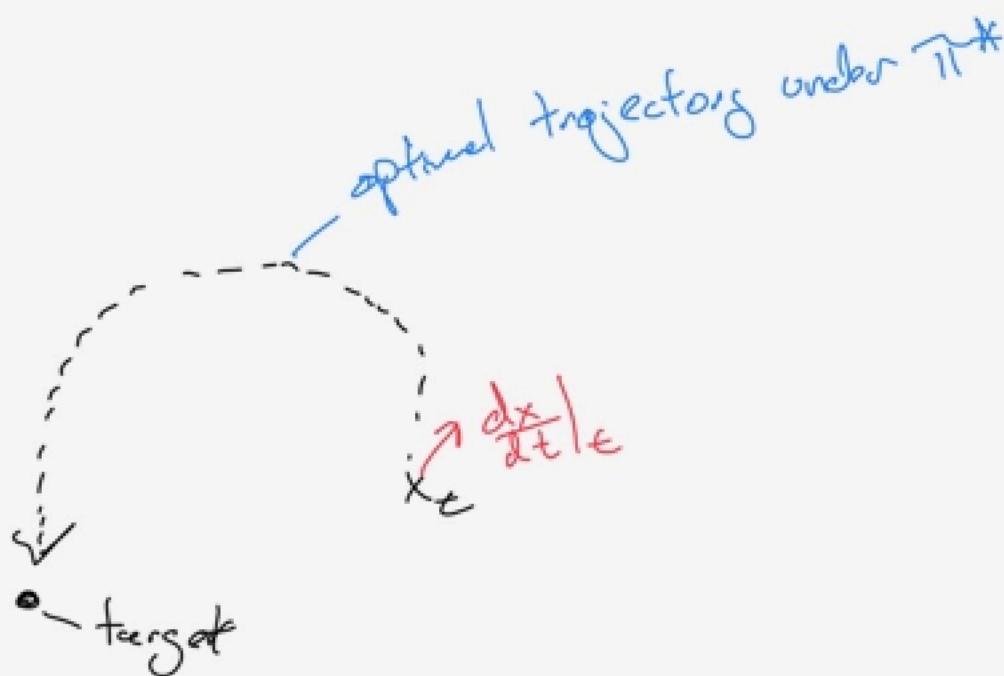
$$\varepsilon \sim N(0, I\sigma)$$

$$A = I$$

B : constrained to a rotation ($\det B = 1$)

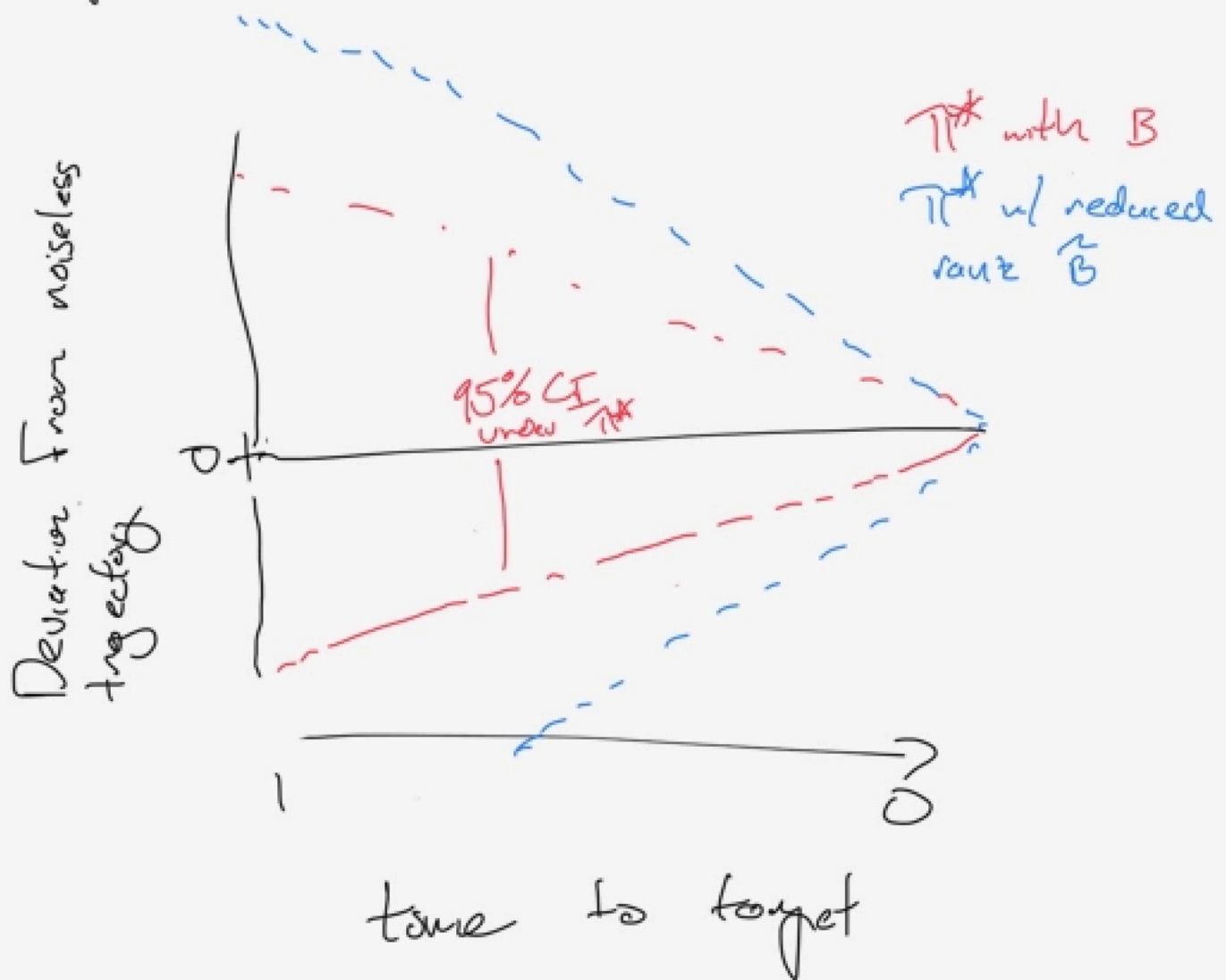
→ This is meant to be a momentum conserving system

e.g



plot 1

→ you can create 1-D plot of the predicted state locat.- under π^*

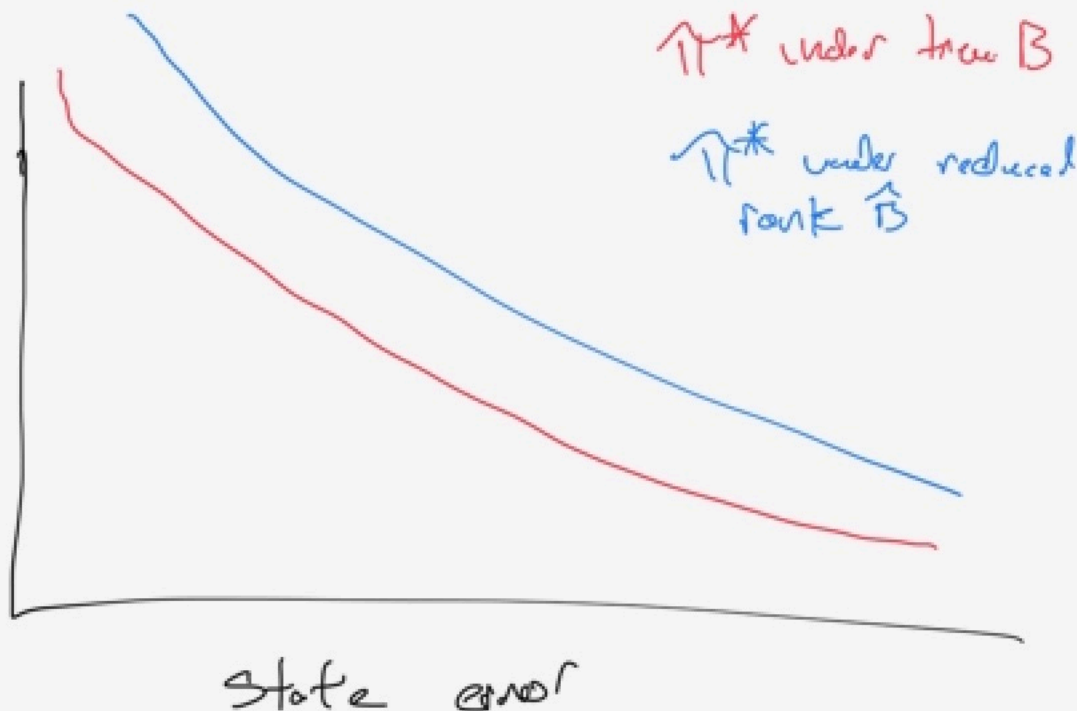


The point of this plot is to show that the forecast error is dominated by the time to target (or approximation error) in the model system. This suggests MPC, even w/ a bad model, is helpful.

Plot Z

$G(\pi)$

Normalized Cumulative
Discounted Reward



$\hat{x}$

$\hat{x}$ $\frac{dx}{d\epsilon}|_+$

$\hat{x}$

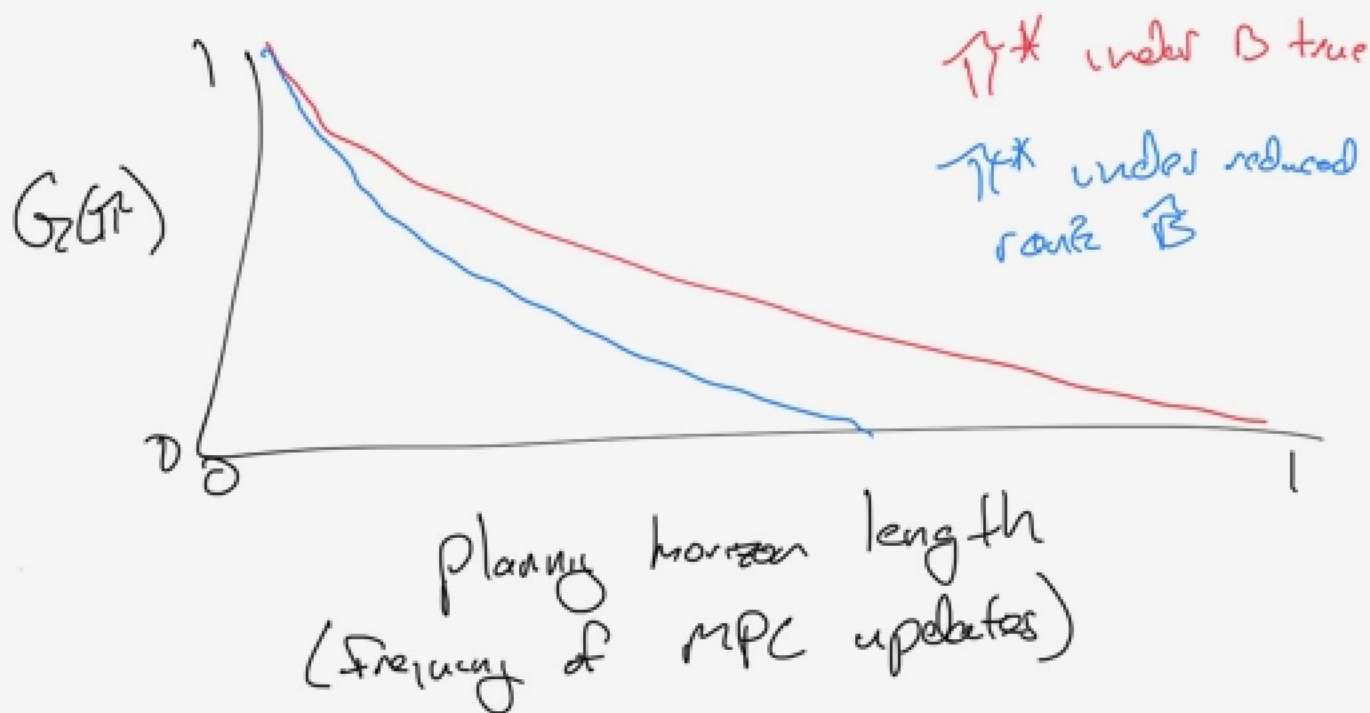
$\hat{x}$

• target

Here, we show that given a noisy state estimate, what are the Expected Returns under the policy? In the model system, state estimation error is higher the further out you predict $\Rightarrow$ MPC reduces the planning horizon.

This plot links the state estimation error to subsequent performance, showing that more frequent updates, even from a bad model, is preferable than long-horizon planning w/ the perfect model.

Plot 3



$G_2(T)$: normalized expected future cost
discounted rewards

This plot completes the argument. As your planning horizon shrinks, the difference in model quality matters less.